The truncated weighted sum of the scale-invariant process is exponentially distributed

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Editorial note. The counted statement is the author's labelled prop environment at source0.tex lines 814-817; its proof environment opens at line 819, the line after the blank separator, so the pairing is by adjacency and the retained proof range 819-821 is the whole environment.

Editorial note. Campbell's theorem prel:campbell (source lines 1815-1837) is retained as context because the counted proof applies it at source line 820; it is the only outside label of the retained proof.

Editorial note. The proof is short and states the computed generating function explicitly rather than only pointing at a citation; the retained range is the whole author proof environment.

Editorial note. This article carries no label-number dump and the retained counter values are left empty for the whole source, so the printed numbering of the card is the wrapper's default; the identity of the counted item is its author label prop:icexponential.

Editorial scope. One original result of the pinned publisher article, counted once: the exponential law of the truncated weighted sum of the scale-invariant process (author label prop:icexponential).

Allowed standard prerequisites (editorial).

- The truncated weighted sum. I_t is the sum of the second coordinates of the points of the scale-invariant process whose first coordinate exceeds t.. It is the random variable whose law is identified.
- Campbell's theorem. For a Poisson process with mean measure mu and a measurable f, the Laplace transform of the sum of f over the points is the exponential of the integral of e to the s f minus one against mu, whenever the integral converges.. It supplies the moment generating function computed in the proof.
- Exponential distribution. A positive random variable with moment generating function t/(t-s) for s below t is exponentially distributed with parameter t.. It converts the computed generating function into the asserted law.

Proposition 0.1 (Campbell's theorem, [11, §3.2]). *Let* Π *be any Poisson process on* S *with mean measure* μ*. Let* $f\colon S \to \mathbb{R}$ *be a measurable function. Then*

$$\Sigma = \sum_{X \in \Pi} f(X)$$

is absolutely convergent almost surely if, and only if,

$$\int_S \min\{f, 1\}\, \mathrm{d}\mu < \infty.$$

If this condition holds, then

$$\mathbb{E}[e^{s\Sigma}] = \exp\left(\int_S (e^{sf} - 1)\, \mathrm{d}\mu\right)$$

for any complex s *for which the integral converges. Moreover,*

$$\mathbb{E}[\Sigma] = \int_S f\, \mathrm{d}\mu$$

if the integral converges. In the case where it converges, we also have

$$\mathrm{Var}[\Sigma] = \int_S f^2\, \mathrm{d}\mu.$$

Proposition 0.2 (Arratia, [2]). *For any fixed* $t > 0$*, the random variable* I_t *follows an exponential distribution of parameter* t*, i.e.* $\mathbb{P}[I_t > y] = e^{-ty}$ *for all* $y > 0$*.*

Proof. A direct application of Campbell's theorem (Proposition 0.1) implies that $\mathbb{E}[e^{sI_t}] = t/(t-s)$ for $\mathrm{Re}(s) < t$, which agrees with the moment generating function of an exponential distribution of parameter t. This completes the proof. □

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